

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Solving Inequalities

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

SOLVING INEQUALITIES · CH2 · L11



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The Map

This lesson collapses Solving Inequalities into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Name the target

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Write the first line

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Calculate carefully

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Answer in the requested form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Solve inequality

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Factorise

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 22 website questions, 22 checked solutions, 3 local PDFs read.

01 Name the target

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Inequalities question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{4-3x}{5} - \frac{3x-5}{2} = -3$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q19

(a)

$$\frac{4-3x}{5} - \frac{3x-5}{2} = -3$$

Multiply by 10

$$2(4-3x) - 5(3x-5) = -30$$

Finish: $x = 3$, and $-\frac{8}{5} \leq y \leq 5$.

FORMULA BANK

Solve like an equation, but flip the sign when multiplying/dividing by a negative.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

19 (a) Solve $4^3 5^3 5^2 3^{---} = -x^x$ Show clear algebraic working. $x = (3)$ (b) Solve the inequality $5y^2 - 17y < 40$ (3)

YOUR SOLUTION

02 Write the first line

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Inequalities question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$-2 \leq y < 1$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1HR Q3

Solve inequality

$$-2 \leq y < 1$$

use a closed circle at

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: Closed at -2 , open at 1 ; integers $-3, -2, -1, 0, 1, 2$.

FORMULA BANK

Solve like an equation, but flip the sign when multiplying/dividing by a negative.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 02.

QUESTION

3 (a) On the number line, show the inequality $-2 < y < 1$ (2) (b) Write down all the values of n that satisfy $-3.4 < n < 2$ (2)

YOUR SOLUTION

03 Calculate carefully

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Inequalities question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4x(2x + 5) - 3x(2x - 3) = 8x^2 + 20x - 6x^2 + 9x = 2x^2 + 29x$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2H Q5

(a)

$$4x(2x + 5) - 3x(2x - 3) = 8x^2 + 20x - 6x^2 + 9x = 2x^2 + 29x$$

Expand brackets

$$\frac{y^5 \times y^n}{y^6} = y^{n-1} = y^{13}$$

Finish: (a) $2x^2 + 29x$, (b) $n = 14$, (c) $t < 3$

FORMULA BANK

Solve like an equation, but flip the sign when multiplying/dividing by a negative.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

5 (a) Expand and simplify $4x(2x + 5) - 3x(2x - 3)$ (2) Given that $y \times y \times y \times n \times 5 \times 6$ times $= y^{13}$
 (b) work out the value of n . $n = (2)$ (c) (i) Solve the inequality $7t - 8 < 2t + 7$ (2) (ii) On
 the number line below, represent the solution set of the inequality solved in part (c)(i)
 $-5 -4 -3 -2 -1 0 1 2 3 4 5 t$ (1)

YOUR SOLUTION

04 Answer in the requested form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Inequalities question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$-2 \leq n < 3$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1HR Q1

(a)

$$-2 \leq n < 3$$

Calculate value

$$n = -2, -1, 0, 1, 2$$

Finish: $n = -2, -1, 0, 1, 2$; closed at 1, shaded left.

FORMULA BANK

Solve like an equation, but flip the sign when multiplying/dividing by a negative.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

1 n is an integer. (a) Write down all the values of n such that $-2 < n < 3$ (2) (b) On the number line, represent the inequality $y < 1$ -3 -2 -1 0 1 2 3 y (1)

YOUR SOLUTION

05 Solve inequality

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Inequalities question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$-2 \leq y < 1$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1HR Q3

Solve inequality

$$-2 \leq y < 1$$

use a closed circle at

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: Closed at -2 , open at 1 ; integers $-3, -2, -1, 0, 1, 2$.

FORMULA BANK

Solve like an equation, but flip the sign when multiplying/dividing by a negative.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 05.

QUESTION

3 (a) On the number line, show the inequality $-2 < y < 1$ (2) (b) Write down all the values of n that satisfy $-3.4 < n < 2$ (2)

YOUR SOLUTION

06 Factorise

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Inequalities question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{4-3x}{5} - \frac{3x-5}{2} = -3$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
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WORKED MODEL

Source pattern: Jan 2020 P2HR Q19

(a)

$$\frac{4-3x}{5} - \frac{3x-5}{2} = -3$$

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Finish: $x = 3$, and $-\frac{8}{5} \leq y \leq 5$.

FORMULA BANK

Solve like an equation, but flip the sign when multiplying/dividing by a negative.

— BOTTOM LINE

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Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Name the target
"work out", "show", "hence", a familiar exam phrase	Write the first line
"work out", "show", "hence", a familiar exam phrase	Calculate carefully
"work out", "show", "hence", a familiar exam phrase	Answer in the requested form
"work out", "show", "hence", a familiar exam phrase	Solve inequality
"work out", "show", "hence", a familiar exam phrase	Factorise

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.